

BUSINESS REQUIREMENTS DOCUMENT (BRD)

1. Business Objective

- The business needs a performance tracking report for advertising campaigns running on Facebook and Instagram.
- The report will provide visibility into campaign reach, engagement, conversions, and budget utilization.
- This will enable the marketing team to:
 - + Identify the most effective platform (Facebook vs Instagram).
 - + Track campaign ROI and optimize budget allocation.
 - + Understand audience engagement patterns.

2. Scope of the Report

- In Scope:
Campaigns running on Facebook and Instagram only.
- Out of Scope:
 - + Other platforms (Messenger, Audience Network).
 - + Organic engagement (only paid ads will be included).

3. KPIs & Definitions

KPI	Definition	Formula	Example use
Impressions	Number of times ads were displayed.	Count of event_type = Impression	Measure reach
Clicks	Number of times users clicked ads.	Count of event_type = Click	Measure engagement intent
Shares	Number of times ads were shared.	Count of event_type = Share	Viral engagement
Comments	Number of user comments on ads.	Count of event_type = Comment	User sentiment & feedback
Purchases	Number of purchases made after seeing ads.	Count of event_type = Purchase	Conversions

Engagements	Total interactions (Clicks + Shares + Comments).	Clicks + Shares + Comments	Engagement volume
CTR (Click Through Rate)	% of impressions that resulted in clicks.	$(\text{Clicks} \div \text{Impressions}) \times 100$	Ad effectiveness
Engagement Rate	% of impressions that resulted in engagements.	$(\text{Engagements} \div \text{Impressions}) \times 100$	Overall ad appeal
Conversion Rate	% of clicks that resulted in purchases.	$(\text{Purchases} \div \text{Clicks}) \times 100$	Funnel efficiency
Purchase Rate	% of impressions that resulted in purchases.	$(\text{Purchases} \div \text{Impressions}) \times 100$	Conversion from reach
Total Budget	Total spend allocated to campaigns.	$\text{Sum of Campaigns} \times \text{total_budget}$	Cost analysis
Avg. Budget per Campaign	Average budget allocation per campaign.	$\text{Total Budget} \div \text{Campaign Count}$	Budget distribution

4. Charts Requirements:

4.1. Target Gender – Donut Chart

A donut chart will visualize performance split by target gender (from the ads table).

- The metric displayed (e.g., Impressions, Clicks, Purchases) will change dynamically via the parameter.
- Purpose: Identify which gender segment contributes most to the selected metric.

4.2. Target Age Group – Bar Chart

A bar chart will show engagement across age groups defined in the ads table.

- Each bar will represent one age group.
- The metric displayed will switch dynamically.
- Purpose: Highlight which age group is most responsive to campaigns.

4.3. Country – Map

A map visualization will display performance by country (from the users table).

- Bubble size or color intensity will represent the selected metric.
- Purpose: Provide a geographic view of campaign reach and engagement.

4.4. Calendar Month – Calendar Heat Map

A calendar heat map will plot performance at the monthly level, based on the timestamp field in ad_events.

- Darker shades will indicate higher activity.
- Purpose: Detect seasonal trends, peak ad months, and low-activity periods.

4.5. Weekly Trend – Stacked Column by Ad Type

A stacked column chart will display weekly performance trends.

- X-axis → Week number (from the Date Table linked to ad_events).
- Stacks → Different ad_type values (from the ads table).
- Y-axis → Selected metric.
- Purpose: Compare ad type contributions over weeks.

4.6. Hourly Trend – Area Chart

An area chart will show activity by hour of day (from ad_events[time_of_day]).

- X-axis → Hour of the day (0–23).
- Y-axis → Selected metric.
- Purpose: Understand user activity patterns throughout the day.

4.7. Ad Type – Matrix

A matrix visualization will show the selected metric across ad types and possibly break down further by platform (Facebook vs Instagram).

- Rows → Ad Types.
- Columns → Platforms or other campaign dimensions.
- Values → Selected metric.
- Purpose: Compare performance across ad formats and platforms side by side.